Labwork 2.2 - Map Operations

General Information and Submission Rules:

- In this labwork, you need to design a map helper for a student to find the desired location in the university.

- By the end of the labwork, you must submit it to Yulearn as **studentID_fullname_labwork2.c**. In addition to this, you need to create a command line history file with this command:
  >> history > history_fullname_labwork1.txts
  and submit this also.
- Incorrect naming or file format will result in <u>a loss of 5 points</u> from your total grade.
- Only use concepts covered in the labs so far.

Question 1 - Calculate Distance

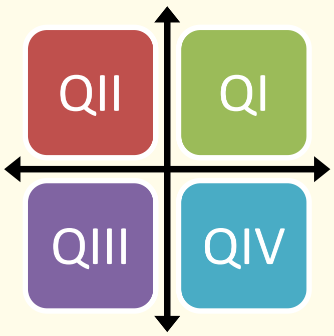
- Create a function (**distance**) that computes the distance between two (x,y) coordinate points. Ensure that the distance should be displayed as only two digits after the decimal point.
- **For here only**, you can use the **pow(base, exponent)** function in the **math.h** library.
- The formula for calculating distance is:

$$distance = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

```
In main:
>> Enter first coordinate: 4 8
>> Enter second coordinate: -9 6
The distance between them is 13.15
```

Question 2 - Find Out Region

- Write a function (**detect_region**) which determines the quadrant of the given point.
- It takes one (x, y) point as a parameter.




```
In main:
>> Enter the first coordinate: 4 8
>> Enter second coordinate: -9 6
The distance between them is 13.15
First is in region Q1.
Second is in region Q2.
```

Question 3 - Draw Points (Loops)

- The function (**draw_points**) draws the representation of a distance function.
- Represent the point name as characters like **A** and **B** (those point names will come as an input from the user) and the distance can be shown with **asterisk** (*).

```
In main:
>> Enter first coordinate: 4 8
>> Enter second coordinate: -9 6
The distance between them is 13.15.
First is in region Q1.
Second is in region Q2.
A ************* B
```

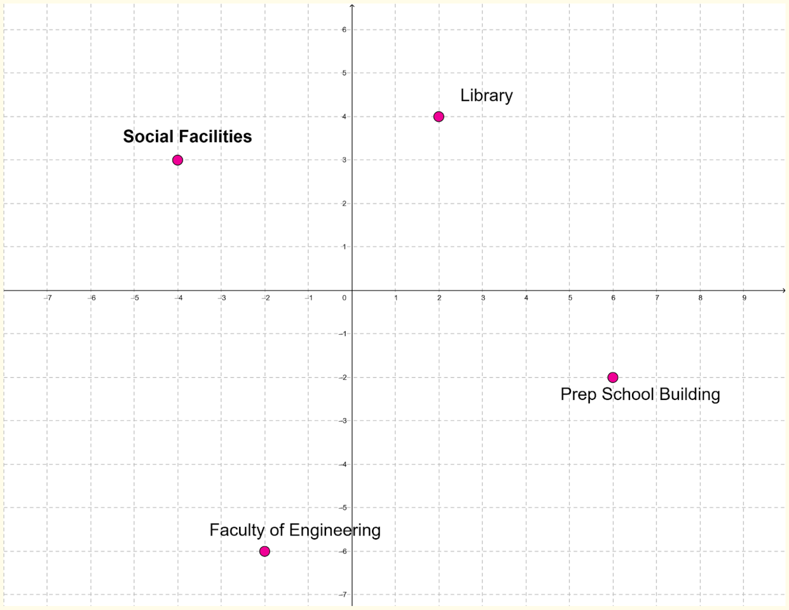
Question 4 - Direction Determination

- The function (**direction**) takes two points as a parameter.
- The first point will be our location and the second is our destination.
- This function will indicate direction.
- For example, if the destination is in the northwest, it will print "go towards the northwest".

```
In main:
>> Enter first coordinate: 4 8
>> Enter second coordinate: -9 6
The distance between them is 13.15.
First is in region Q1.
Second is in region Q2.
A ************* B
Please go towards the southwest!
```

Question 5 - Student Map

- Use created functions to do given operations.
- The buildings in the university are represented in the coordinate plane as given points:
  Library: A (2, 4)
  Social Facilities: B (-4, 3)
  Faculty of Engineering: C (-2, -6)
  Prep School Building: D (6, -2)



- Create a function (**student_map**) with given properties:
    - Display all point's quadrants to the user in the beginning.
    - Take two point names from the students like A and B.
    - Then, print out the total distance between those locations. It will be in terms of steps.
    - Draw the distance and tell which way the student needs to go.
    - Assume each step is equivalent to 0.5 calories, return the value that the student is going to burn.

```
In main:
A is in region Q1.
B is in region Q2.
C is in region Q3.
D is in region Q4.
>> Enter your location: A
>> Enter destination: B
The distance between them is 6.08 steps.
A ****** B
Please go towards the southwest!
You will burn 3 calories.
```